W-B-133D
July 26, 1985
SUPERSEDING
W-B-133C
May 11, 1979

FEDERAL SPECIFICATION

BATTERY, STORAGE
(LEAD-ACID, INDUSTRIAL PORTABLE SERVICE)

This specification is approved by the Assistant Administrator, Office of Federal Supply and Services, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers industrial batteries of the lead-acid type for use in electric fork trucks, electric tractors, and other comparable applications.

1.2 Classification. Individual storage batteries, as covered by this specification, shall be of the voltage, ampere-hour capacity, and within the the dimensional limitations, as specified in the invitation for bids.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on the date of the invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specifications:

O-S-801 - Sulfuric Acid, Electrolyte; for Storage Batteries.
PPP-B-140 - Batteries, Storage, Industrial, Packaging of

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards, and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

FSC 6140

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

(Federal Government activities may obtain copies of Federal standardization documents and the Index of Federal Specifications, Standards, and Commercial Item Descriptions from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Assn., Inc., Traffic Department, 1616 P Street, N. W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

American Society for Testing and Materials (ASTM) Publications:

D 639 - Battery Containers Made from Hard Rubber or Equivalent Materials.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Electrical Manufacturers Assn. (NEMA) Publications:

IB 1 - Definitions and Precautionary Labels for Lead-Acid Industrial Batteries.

IB 2 - Determination of Capacity of Lead-Acid Industrial Storage Batteries for Motive Power Service.

IB 3 - Cycle Life Testing of Lead-Acid Industrial Storage Batteries for Motive Power Service.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

(Application for copies should be addressed to the National Electrical Manufacturers Assn., 2101 L Street, N.W., Suite 300, Washington, DC 20037.)

3. REQUIREMENTS

3.1 General. Battery components and materials shall be in accordance with the manufacturer's standard commercial practice.

3.2 Definitions and precautions. The definitions and precautionary labels shall be in accordance with NEMA Standard IB 1.

3.3 Cell.

3.3.1 Element. The design of the element shall ensure that the cell meets the specified electrical requirements.

3.3.2 Grids. The grids shall be made of a suitable lead alloy.

3.4 Electrolyte. The nominal gravity of the electrolyte shall be within the range shown on the battery nameplate or in the manufacturer's published instructions. The electrolyte shall be in accordance with O-S-801. Except for organic matter, the values of the impurities shall be determined by any recognized analytical test method. The organic matter shall be determined in accordance with paragraph 4.4.8.1.

3.5 Terminal markings. The polarity of the positive terminal shall be permanently marked "POS", "P", or "+". The negative terminal shall be marked "NEG", "N", or "-". In addition, colors may be used on the terminals; red for positive and black for negative.

3.6 Connectors. Cable, cell, and terminal connectors, other than exposed lead parts, shall be protected against corrosion. The type of charging connector, position, and length of leads shall be as specified (see 6.1).

3.7 Jars. Jars shall be of single cell or multi-cell construction. The material shall be commercially acceptable and free of rough spots, pinholes, or cracks. When subjected to the electrical breakdown test in 4.4.7, there shall be no evidence of perforation or burn through.

3.7.1 Impact resistance. The impact resistance of sample pieces cut from single cell and multi-cell jars shall be 120 inch-pounds minimum, when tested in accordance with 4.4.5.

3.7.2 Sediment space. Jars shall have sufficient sediment space to minimize contact between the sediment and element that may cause a short.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the current version before use.

3.7.3 Space above element. The space above the element shall be in accordance with commercial practice.

3.7.4 Acid-resistance. The jars and covers shall not absorb an excessive amount of electrolyte, when tested in accordance with 4.4.6. The material shall not comply if the increase in weight exceeds 1.5 percent, if visible blisters or cracks are found, or if the dimensions are increased 2 percent or more.

3.8 Cell covers. The cell cover shall be compatible with the jar. The design of the cover shall prevent distortion of the cover and leakage of electrolyte when the cell is operated in accordance with the manufacturer's instructions.

3.8.1 Vent caps. Suitable baffling shall be provided in the cell or vent to avoid spraying and minimize loss of electrolyte while permitting gas to escape during charging, discharging, or open circuit.

3.8.2 Seal. The joint between the cover and jar shall be gas-tight and effectively seal against leakage of electrolyte. An unbroken seal between the cover and jar shall be maintained throughout a temperature range of 0 degrees F (-17 degrees C) to +140 degrees F (+60 degrees C).

3.9 Trays. Trays, when furnished, shall be of suitable strength, construction, and material, for the intended service and equal to the manufacturer's standard product. All surfaces of metal covers and lifting means and all exposed vertical surfaces of metal trays shall be protected against exposure to sulfuric acid.

3.10 Assembly.

3.10.1 Cells. Cells shall be furnished as single cells, trays of cells, or complete batteries, as specified in the invitation for bid or order.

3.10.2 Batteries. Unless otherwise specified, the battery shall be of standard groupings, dimensions, and connections and include a charging plug, receptacle, or suitable terminal connection. Batteries shall conform to dimensional limitations specified (see 1.2). Cell connectors shall be burned (fused together) or bolted to the terminals and be adequate to carry the rated current.

3.11 Weights. The actual weight of the trayed battery shall be permanently stamped into the wall of the tray near one of the lifting means as "SERVICE WEIGHT - ___ LBS" in upper case letters. This weight shall be within ± 5 percent of the manufacturer's published catalog value.

3.12 Charge condition. As specified in the solicitation (see 6.1), cells or batteries shall be shipped either charged-and-wet or charged-and-dry. When the purchaser does not specify , batteries shall be shipped charged-and-wet and ready for service.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

3.12.1 Charged-and-wet. Batteries shall be shipped filled with electrolyte of proper specific gravity, fully charged and ready for use.

3.12.2 Charged-and-dry. Batteries intended for storage in the charged-and-dry condition within the time limit specified by the manufacturer, shall contain a dry element. When necessary to prevent deterioration of the element due to exposure to moisture, vents shall be sealed. When filled with electrolyte, according to the manufacturer's instructions, batteries shall be required to deliver 90 percent of their rated capacity on or before the third cycle when discharged as specified in 4.4.3.

3.13 Markings. Batteries shall be legibly and permanently marked with the following:

(a) Name or trademark of the manufacturer.
(b) Manufacturer's designation.
(c) Six-hour ampere capacity rating in ampere hours.
(d) Month and year of manufacture.
(e) Service weight of trayed batteries.
(f) Nominal gravity.
(g) Warning labels.

3.14 Instructions. Each battery shall have a set of instructions for the proper installation and maintenance of the battery, including cycling characteristics. Pertinent data shall identify the battery with relation to type, size, and manufacturer. The instructions shall be securely fastened and protected from mechanical and chemical damage.

3.15 Performance.

3.15.1 Rated capacity. After conditioning in accordance with 4.4.2, cells or batteries shall have not less than 90 percent of their rated ampere-hour capacity (see 1.2), at the end of the third cycle, when tested in accordance with 4.4.3. The rated capacity shall not exceed the manufacturer's commercial rating. The number of cycles necessary to deliver 100 percent of the manufacturer's rated capacity shall not exceed 10.

3.15.2 Cycle life. Unless otherwise specified (see 6.1), the cell or battery shall complete a minimum of 1500 cycles when tested in accordance with 4.5. The cells or batteries tested shall be representative of the range of capacities of the cells or batteries produced.

3.15.3 Retention of charge. Single cells or cells of a battery shall not show a drop in specific gravity greater than 0.015 units in 14 days, when tested in accordance with 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

3.16 Workmanship. The active material shall be free from breaks and loose parts. The grids shall be free of open sections, cold pouring, cracks, and blowholes. Breaks are permitted in those parts of the grid used to retain the active material only when the break does not impair the strength of the grid or permit the active material to become separated from the plate. The sections of the element involving lead burning shall be free from defective bonds.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all the inspection requirements specified herein.

4.2 Lot. Submission of the product and the formation of lots shall be in accordance with MIL-STD-105.

4.2.1 Sampling for inspection. Sampling for inspection shall be in accordance with MIL-STD-105, level S-1.

4.2.2 Classification of defects. For inspection purposes, defects shall be classified as follows:

Major defects (1.0 AQL)
- Elements (see 3.3.1)
- Grids (see 3.3.2)
- Electrolyte (see 3.4)
- Terminal markings (see 3.5)
- Connectors (see 3.6)
- Jars (see 3.7, 3.7.1, 3.7.2, 3.7.3, and 3.7.4)
- Cell covers (see 3.8, 3.8.1, and 3.8.2)
- Trays (see 3.9)
- Assembly (see 3.10.1 and 3.10.2)
- Weights (see 3.11)
- Charge condition (see 3.12, 3.12.1, and 3.12.2)
- Markings (see 3.13)
- Workmanship (see 3.16)
- Dimensions (see Appendix 1 and 6.1c)

Minor defects (2.5 AQL)
- Missing instructions

4.3 Sampling for tests. Sampling for tests shall be in accordance with MIL-STD-105, inspection level S-1 with an AQL of 1.0 percent defective. The lot size shall be designated as all of the single cells or batteries of one design produced in one calendar year at one location. The sample size shall be 13 cells or batteries per design for test per year. Sample cells shall be of the 11-plate size or larger, and shall be considered as representative of the range of cell sizes produced. Periodic selection of samples for testing shall be made at intervals not exceeding four months.

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

4.4 Tests.

4.4.1 Verification. Verification of tests shall be determined by review of test records and data. Results of prior testing of cells or batteries selected in accordance with 4.3 and tested in accordance with 4.4.3, capacity test; 4.4.4, retention of charge test; and 4.5, cycle life shall be acceptable to demonstrate conformance of batteries being furnished on current orders or contracts.

4.4.2 Conditioning of samples. If supplied in a charged-and-wet condition, the cell or battery may be brought to a fully charged state before discharging. If supplied in a charged-and-dry condition, the battery shall be initially conditioned in accordance with the manufacturer's instructions.

4.4.3 Capacity test. To determine compliance with 3.15.1, cells or batteries shall be subjected to capacity tests in accordance with NEMA Standard IB 2.

4.4.4 Retention of charge test. To determine conformance with 3.15.3, the cells or batteries shall be fully charged with the specific gravity of each cell corrected to 77 degrees F (25 degrees C). The cell or battery shall then be allowed to stand on open circuit in an ambient temperature of 75 to 90 degrees F (23.8 to 32.2 degrees C) for a period of 14 days. At the end of the 14-day period, the specific gravity shall be measured and corrected to 77 degrees F (25 degrees C).

4.4.5 Impact resistance. To determine comformance with 3.7.1, samples from jars shall be tested in accordance with ASTM D 639. The size of the specimen to be tested on the apparatus shown in figure 1 of ASTM D 639 shall be of sufficient size. The area of impact shall be the geometric center of the test specimen.

4.4.6 Acid-resistance test. To determine conformance to 3.7.4, jars and covers shall be tested in accordance with ASTM D 639.

4.4.7 Electrical breakdown test. To determine conformance with 3.7, jars shall be tested in accordance with ASTM D 639. The voltage applied shall be 100 volts per mil of wall thickness.

4.4.8 Electrolyte. The impurity values specified in O-S-801 (except organic matter) shall be determined by any recognized analytical test method (see 3.4).

4.4.8.1 Organic matter. Heat 200 ml of electrolyte in a clean beaker until the acid begins to fume strongly. To pass the test, there shall be no perceptible charring; a slight yellow color that appears when the acid is hot but disappears on cooling should be disregarded.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

4.5 Cycle life. To determine conformance with 3.15.2, the cycle shall be determined in accordance with NEMA Standard IB 3. The cells shall be cycled two times a day but not less than 10 cycles per week. Note: Verification shall be determined by review of acceptable test records and data showing compliance with the cycle requirement.

4.6 Examination of preparation for delivery. An examination shall be made to determine compliance with the requirements of section 5. The sample unit shall be one shipping container fully prepared for delivery. Sampling shall be in accordance with MIL-STD-105. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of percent defective.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or commercial, as specified (see 6.1).

5.1.1 Level A. Each battery shall be preserved and packaged an accordance with level A requirements of PPP-B-140.

5.1.2 Commercial. Each battery shall be preserved and packaged in accordance with normal commercial practice. The complete package shall be designed to protect the battery against damage during shipment, handling, and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.1).

5.2.1 Level A. Batteries of like description, packaged as specified in 5.1.1, shall be packed in accordance with the level A requirements of PPP-B-140.

5.2.2 Commercial. Batteries of like description, packaged as specified in 5.1.2, shall be packed in containers that will assure acceptance by common carrier and provide product protection against loss and damage during multiple shipments, handling, and storage. The shipping container shall be in compliance with the National Motor Freight Classification and Uniform Freight Classification.

5.3 Marking. Marking shall be as specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026l0-26T11:25Z
Check the source to verify that this is the currentversion before use.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of battery required (see 1.1).
(c) Voltage, ampere-hour capacity, number of cells, and overall dimensions (see 1.2).
(d) Battery components, if other than specified (see 3.1).
(e) Specific gravity of cells (see 3.4).
(f) Tray material and whether cover is required (see 3.9).
(g) Battery assembly (see 3.10, 3.10.1 and 3.10.2).
(h) Condition of charge (see 3.12).
(i) Cycle life, if other than specified (3.15.2).
(j) Type of charging connector, position, and length of leads (see3.6).
(k) Inspection responsibility, if other than specified (see 4.1).
(l) Selection of applicable levels of preservation and packaging (see 5.1) and packing (see 5.2).

MILITARY INTERESTS:

User Activity

Army - ER

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - PCD
TVA

PREPARING ACTIVITY:

GSA - FSS

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

APPENDIX I

W-B-133 INDUSTRIAL BATTERY CHARACTERISTICS

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
104-0964	36	Dwe	85	965	12	6	19		38.4x7.8x23.5
106-0101	88	Dwe	145	1015	12	6	15		36.1x9.1x31.2
106-0102	62	Dwe	100	500	48	24	11		38.1x18x31.2
107-0190	4	Dwe	75	375	12	6	11		26.3x7.7x23
107-0191	8	Dwe	75	675	12	6	19		22.7x13x13.5
107-0192	16	Dwe	75	675	32	16	19		30.4x25.7x23.5
107-0193	24	Dwe	75	825	36	18	23		38.5x27.3x23.5
107-0194	40	Dwe	90	540	12	6	13		31.2x7.7x26.5
107-0195	48	Dwe	90	810	36	18	19		38.4x22.3x25.6
107-0196	72	Dwe	110	660	24	12	13		31.2x12.9x31.2
107-0197	76	Dwe	110	550	36	18	11		30.1x18x31.2
107-0198	80	Dwe	110	440	48	24	9		38.1x15x31.2
107-0199	84	Dwe	110	880	48	24	17		40.7x25.7x31.2
107-0200	95	Dwe	145	580	36	18	9		26.6x22.7x31.2
107-0201	92	Dwe	145	870	24	12	13		35.5x16.1x31.2
107-0202	98	Dwe	145	1015	36	18	15		36.2x26.6x31.2
107-0203	28	Dwe	75	675	48	24	19		38.5x30.4x23.5
107-0204	6	Dwe	75	525	12	6	15		36.0x7.7x23.5
107-0205	5	Dwe	75	450	12	6	13		31.2x7.7x23.0
107-0206	7	Dwe	75	600	12	6	17		20.5x13.0x23.5
107-0207	96	Dwe	145	725	36	18	11		27.2x26.6x31.2
107-0208	97	Dwe	145	870	36	18	13		40.7x21.2x31.2
107-0209	93	Dwe	145	1015	24	12	15		36.2x18.0x31.2
107-0210	99	Dwe	145	1160	36	18	17		41.5x27.0x31.2
107-0211	29	Dwe	75	750	48	24	21		38.5x33.4x23.5
107-0212	30	Dwe	75	825	48	24	23		38.5x36.4x23.5
107-0213	73	Dwe	110	770	24	12	15		36.1x13.0x31.2
107-0214	74	Dwe	110	880	24	12	17		40.6x13.0x31.2
107-0215	75	Dwe	110	440	36	18	9		22.5x19.1x31.2
107-0216	77	Dwe	110	770	36	18	13		31.4x19.2x31.2
108-2221	44	Dwe	90	450	36	18	11		26.12x19.3x25.6
108-2222	52	Dwe	100	700	12	6	15		36.1x7.7x31.2
108-2223	1	Dwe	75	150	12	6	5		12.6x6.6x23.3
108-2224	12	Dwe	75	525	24	12	15		25.8x18.2x23.5
108-2225	60	Dwe	100	700	36	18	15		31.9x24.2x31.2
108-2226	64	Dwe	100	700	48	24	15		36.2x25.7x31.2
108-2227	68	Dwe	110	660	12	6	13		31.2x7.7x31.2
108-2931	20	Dwe	75	525	36	18	15		31.9x24.2x23.5
108-2932	56	Dwe	100	700	24	12	15		25.8x18.2x31.2
108-9653	94	Dwe	145	1160	24	12	17		36.2x18.0x31.2
108-9654	31	Dwe	85	340	12	6	9		22.0x6.6x23.5
108-9655	78	Dwe	110	770	36	18	15		31.9x24.2x31.2
108-9656	61	Dwe	100	800	36	18	17		33.7x25.7x31.2
109-4266	79	Dwe	110	880	36	18	17		38.5x20.6x31.2
109-4267	69	Dwe	110	770	12	6	15		36.1x7.7x31.2
109-4268	70	Dwe	110	440	24	12	9		25.5x11.2x31.2
109-4269	71	Dwe	110	550	24	12	11		31.3x11.0x31.2
109-4270	13	Dwe	75	675	24	12	19		33.4x19.4x23.5

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
109-4271	14	Dwe	75	825	24	12	23		23.4x19.5x23.5
109-4272	15	Dwe	75	600	32	16	17		23.4x25.7x23.5
109-4273	22	Dwe	75	675	36	18	19		36.7x25.7x23.5
109-4274	23	Dwe	75	750	36	18	21		29.7x25.7x23.5
109-4275	89	Dwe	145	1160	12	6	17		46.4x9.2x31.2
109-4276	90	Dwe	145	580	24	12	9		35.3x11.4x31.2
109-4277	91	Dwe	145	725	24	12	11		35.3x13.6x31.2
109-4278	57	Dwe	100	800	24	12	17		27.3x19.3x31.2
109-4279	58	Dwe	100	500	36	18	11		30.1x17.8x31.2
109-4280	59	Dwe	100	600	36	18	13		30.6x20.8x31.2
109-4281	54	Dwe	100	500	24	12	11		32.3x11.0x31.2
109-4282	55	Dwe	100	600	24	12	13		31.2x12.9x31.2
109-4283	49	Dwe	90	900	36	18	21		38.4x24.6x25.6
109-4284	50	Dwe	100	500	12	6	11		22.0x6.6x31.2
109-4285	51	Dwe	100	600	12	6	13		31.2x7.7x23.5
109-4286	41	Dwe	90	630	12	6	15		36.1x7.7x26.5
109-4287	37	Dwe	85	850	12	6	21		25.0x13.0x34.0
109-4288	38	Dwe	85	935	12	6	23		38.4x9.2x23.5
109-4289	39	Dwe	90	450	12	6	11		26.6x7.7x26.5
109-4290	81	Dwe	110	550	48	24	11		38.1x18.0x31.2
109-4291	82	Dwe	110	660	48	24	13		38.1x21.0x31.2
109-4292	83	Dwe	110	770	48	24	15		38.5x24.2x31.2
109-4293	63	Dwe	100	600	48	24	13		31.1x25.7x31.2
109-4294	65	Dwe	100	800	48	24	17		38.5x27.2x31.2
109-4295	9	Dwe	75	750	12	6	21		25.0x13.0x23.5
109-4296	11	Dwe	75	375	24	12	11		26.7x12.9x23.5
109-4297	10	Dwe	75	225	24	12	7		34.4x6.6x23.5
109-4298	2	Dwe	75	225	12	6	7		17.2x6.6x23.3
109-4299	3	Dwe	75	300	12	6	9		22.0x6.6x23.3
110-2060	42	Dwe	90	720	12	6	17		20.4x13.0x26.5
110-2061	43	Dwe	90	810	12	6	19		38.4x7.7x26.5
110-2062	45	Dwe	90	540	36	18	13		30.5x20.1x25.6
110-2063	46	Dwe	90	630	36	18	15		35.3x19.4x25.6
110-2064	47	Dwe	90	720	36	18	17		39.0x19.8x25.6
110-2065	33	Dwe	85	510	12	6	13		31.2x7.8x23.5
110-2066	34	Dwe	85	595	12	6	15		36.1x7.8x23.5
110-2067	35	Dwe	85	680	12	6	17		20.4x13.4x23.5
110-2068	85	Dwe	145	580	12	6	9		26.5x7.7x31.2
110-2069	87	Dwe	145	870	12	6	13		31.6xd9.1x31.2
110-2070	25	Dwe	75	450	48	24	13		31.1x25.7x23.5
110-2071	26	Dwe	75	525	48	24	15		36.2x25.7x23.5
110-2072	27	Dwe	75	600	48	24	17		38.5x27.2x23.5
110-2073	18	Dwe	75	825	32	16	23		36.4x25.7x23.5
110-2074	17	Dwe	75	750	32	16	21		33.4x25.7x23.5
110-2075	19	Dwe	75	450	36	18	13		30.6x20.8x13.5
110-3959	32	Dwe	85	425	12	6	11		26.6x7.8x23.5
110-4455	53	Dwe	100	800	12	6	17		38.4x7.7x31.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

(a) Title, number, and date of this specification.

NSN 6140-01-	Ind No.	Chg **	AH / + PL	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
110-6101	21	Dwe	75	600	36	18	17		33.8x25.7x23.5
113-9577	86	Dwe	145	725	12	6	11		27.0x9.1x31.2
117-1855	117	Dwe	55	385	36	18	15	4x5cc	30.5x26.0x18.6
118-3643	120	Dwe	100	600	36	18	13	3x6	38.2x16.5x30.5
118-3644	113	Dwe	85	680	36	18	17	3x6	34.2x26.5x23.6
118-3645	110	Dwe	75	525	36	18	15	4x5	32.2x24.5x22.6
118-3646	109	Dwe	75	525	36	18	15	3x6	36.5x19.7x22.6
118-3647	108	Dwe	85	510	36	18	13	3x6	31.5x19.6x22.6
118-3648	106	Dwe	120	720	36	18	13	3x6	38.2x16.5x30.5
118-3649	103	Dwe	100	450	24	12	12	2x6	31.7x13.1x23.2
118-3650	100	Dwe	145	870	12	6	13	2x3	32.3x9.2x31.0
118-3651	125	Dwe	85	680	48	24	17	4x6	38.7x27.5x22.6
118-3652	124	Dwe	85	850	48	24	21	4x6	38.7x33.5x22.6
118-3653	118	Dwe	75	825	36	18	23	4x5cc	38.7x27.5x22.6
118-8964	115	Dwe	75	600	36	18	17	4x5cc	32.3x27.6x22.6
118-8965	112	Dwe	75	675	36	18	19	3x6	38.7x23.1x22.6
118-8966	121	Dwe	75	600	48	24	17	4x6	38.7x27.5x22.6
118-8967	122	Dwe	75	675	48	24	19	4x6	38.7x30.5x22.6
118-9943	114	Dwe	75	600	36	18	17	4x5	32.3x27.6x22.6
118-9944	111	Dwe	75	600	36	18	17	3x6	38.7x20.8x22.6
118-9945	107	Dwe	100	600	36	18	13	3x6	31.5x19.6x22.6
118-9946	102	Dwe	75	450	24	12	13	2x6	31.7x13.1x23.2
118-9947	101	Dwe	85	595	12	6	15	2x3	18.7x13.1x23.2
120-0985	123	Dwe	75	750	48	24	21	4x6	38.7x33.5x22.6
120-0986	116	Dwe	75	975	36	18	27	3x6	38.2x32.1x22.6
122-2657	119	Dwe	90	630	36	18	15		32.2x24.5x27.1
124-5398	105	Dwe	120	600	36	18	11	3x6	38.2x14.0x31.0
124-5399	104	Dwe	110	550	36	18	11	3x6	38.2x14.0x31.0
186-8388	2	W	75	225	12	6	7		12.6x6.6x23.3
186-8393	67	Dwe	110	550	12	6	11		26.6x7.7x31.2
186-8394	66	Dwe	110	440	12	6	9		22.0x6.6x31.2
186-8395	2	Dwo	75	225	12	6	7		17.2x6.6x23.3
186-8396	1	W	75	150	12	6	5		12.6x6.6x23.3
186-8397	1	Dwo	75	150	12	6	5		12.6x6.6x23.3
186-8398	5	W	75	450	12	6	13		31.2x7.7x23.0
186-8399	5	Dwo	75	450	12	6	13		31.2x7.7x23.0
186-8400	4	W	75	375	12	6	11		26.3x7.7x23.0
186-8401	4	Dwo	75	375	12	6	11		26.3x7.7x23.0
186-8402	3	Dwo	75	300	12	6	9		22.0x6.6x23.3
186-8403	3	W	75	300	12	6	9		22.0x6.6x23.3
192-2493	6	W	75	525	12	6	15		36.0x7.7x23.5
192-2494	6	Dwo	75	525	12	6	15		36.0x7.7x23.5
192-5552	7	Dwo	75	600	12	6	17		20.5x13.0x23.5
192-5553	8	W	75	675	12	6	19		22.7x13.0x23.5
192-5554	9	W	75	750	12	6	21		25.0x13.0x23.5
192-5555	9	Dwo	75	750	12	6	21		25.0x13.0x23.5
192-5556	10	W	75	225	24	12	7		34.4x6.6x23.5
192-5557	10	Dwo	75	225	24	12	7		34.4x6.6x23.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.

NSN 6140-01-	Ind No.	Chg **	Ah / + PL	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
192-5558	11	W	75	375	24	12	11		26.7x12.9x23.5
192-5559	20	Dwo	75	525	36	18	15		31.9x24.2x23.5
192-5560	21	W	75	600	36	18	17		33.8x25.7x23.5
192-5561	11	Dwo	75	375	24	12	11		26.7x12.9x23.5
192-5562	12	Dwo	75	525	24	12	15		25.8x18.2x23.5
192-5563	13	W	75	675	24	12	19		33.4x19.4x23.5
192-5564	13	Dwo	75	675	24	12	19		33.4x19.4x23.5
192-5565	14	W	75	825	24	12	23		36.4x19.4x23.5
192-5566	15	W	75	600	32	16	17		27.2x25.7x23.5
192-5567	15	Dwo	75	600	32	16	17		27.2x25.7x23.5
192-5568	16	W	75	675	32	16	19		30.4x25.7x23.5
192-5569	16	Dwo	75	675	32	16	19		30.4x25.7x23.5
192-5570	17	Dwo	75	750	32	16	21		33.4x25.7x23.5
192-5571	18	Dwo	75	825	32	16	23		36.4x25.7x23.5
192-6207	22	W	75	675	36	18	19		36.7x25.7x23.5
192-6208	22	Dwo	75	675	36	18	19		36.7x25.7x23.5
192-6209	23	W	75	750	36	18	21		39.7x25.7x23.5
192-6210	24	W	75	825	36	18	23		38.5x27.7x23.5
192-6211	24	Dwo	75	825	36	18	23		38.5x27.7x23.5
192-6212	25	W	75	450	48	24	13		31.1x25.7x23.5
192-6213	25	Dwo	75	450	48	24	13		31.1x25.7x23.5
192-6214	26	W	75	525	48	24	15		36.2x25.7x23.5
192-6215	26	Dwo	75	525	48	24	15		36.2x25.7x23.5
192-6216	27	Dwo	75	600	48	24	17		38.5x27.2x23.5
192-6217	28	Dwo	75	675	48	24	19		38.5x30.4x23.5
192-6218	29	Dwo	75	750	48	24	21		38.5x33.4x23.5
192-6219	30	W	75	825	48	24	23		38.5x36.4x23.5
192-6220	30	Dwo	75	825	48	24	23		38.5x36.4x23.5
192-6221	31	W	85	340	12	6	9		22.0x6.6x23.5
192-6222	31	Dwo	85	340	12	6	9		22.0x6.6x23.5
192-6223	32	W	85	375	12	6	11		26.6x7.8x23.5
192-6224	32	Dwo	85	375	12	6	11		26.6x7.8x23.5
192-6225	33	W	85	450	12	6	13		31.2x7.8x23.5
192-6226	110	W	75	525	36	18	15	4x5	32.2x24.5x22.6
192-6227	111	W	75	600	36	18	17	3x6	38.7x20.8x22.6
192-6228	112	W	75	675	36	18	19	3x6	38.7x23.1x22.6
192-6229	114	W	75	600	36	18	17	4x5	32.3x27.6x22.6
192-6230	115	W	75	600	36	18	17	4x5cc	32.3x27.6x22.6
192-6231	116	W	75	975	36	18	27	3x6	38.2x32.1x22.6
192-6232	117	W	55	385	36	18	15	4x5cc	30.5x26.0x18.6
192-6233	118	W	75	825	36	18	23	4x5cc	38.7x27.6x
192-6234	119	W	90	630	36	18	15		32.2x24.5x27.1
192-6235	121	W	75	600	48	24	17	4x6	38.7x27.5x22.6
192-6236	122	W	75	675	48	24	19	4x6	38.7x30.5x22.6
192-6237	123	W	75	750	48	24	21	4x6	38.3x33.5x22.6
192-6238	125	W	85	680	48	24	17	4x6	38.7x27.5x22.6
192-6239	33	W	85	510	12	6	13		31.2x7.8x23.5

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the current version before use.

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
192-6240	95	Dwo	145	580	36	18	9		26.6x22.7x31.2
192-6241	96	Dwo	145	725	36	18	11		27.2x26.6x31.2
192-6242	96	W	145	725	36	18	11		27.2x26.6x31.2
192-6243	97	Dwo	145	870	36	18	13		40.7x21.2x31.2
192-6244	97	W	145	870	36	18	13		40.7x21.2x31.2
192-6245	98	Dwo	145	1015	36	18	15		36.2x26.6x31.2
192-6246	98	W	145	1015	36	18	15		36.2x26.6x31.2
192-6247	99	Dwo	145	1160	36	18	17		41.5x27.0x31.2
192-6248	102	W	75	450	24	12	13	2x6	31.7x13.1x23.2
192-6249	103	W	100	700	24	12	15	2x6	36.2x13.1x31.0
192-6250	104	W	110	550	36	18	11	3x6	38.2x14.0x31.0
192-6251	107	W	100	600	36	18	13	3x6	31.5x19.6x22.6
192-6252	108	W	85	510	36	18	13	3x6	31.5x19.6x22.6
192-6253	109	W	75	525	36	18	15	3x6	36.5x19.7x22.6
192-6583	59	Dwo	100	600	36	18	13		30.6x20.8x31.2
192-6584	60	Dwo	100	700	36	18	15		31.9x24.2x31.2
192-6585	60	W	100	700	36	18	15		31.9x24.2x31.2
192-6586	61	Dwo	100	800	36	18	17		33.7x25.7x31.2
192-6587	61	W	100	800	36	18	17		33.7x25.7x31.2
192-6588	62	Dwo	100	500	48	24	11		38.1x18.0x31.2
192-6589	62	W	100	500	48	24	11		38.1x18.0x31.2
192-6590	63	Dwo	100	600	48	24	13		31.1x25.7x31.2
192-6591	63	W	100	600	48	24	13		31.1x25.7x31.2
192-6592	64	Dwo	100	700	48	24	15		36.2x25.7x31.2
192-6593	64	W	100	700	48	24	15		36.2x25.7x31.2
192-6594	65	Dwo	100	800	48	24	17		38.5x27.2x31.2
192-6595	66	Dwo	110	440	12	6	9		22.0x6.6x31.2
192-6596	66	W	110	440	12	6	9		22.0x6.6x31.2
192-6597	67	Dwo	110	550	12	6	11		26.6x7.7x31.2
192-6598	67	W	110	550	12	6	11		26.6x7.7x31.2
192-6599	68	Dwo	110	660	12	6	13		31.2x7.7x31.2
192-6600	68	W	110	660	12	6	13		31.2x7.7x31.2
192-6601	69	Dwo	110	770	12	6	15		36.1x7.7x31.2
192-6602	69	W	110	770	12	6	15		36.1x7.7x31.2
192-6603	70	Dwo	110	440	24	12	9		25.5x11.2x31.2
192-6604	70	W	110	440	24	12	9		25.5x11.2x31.2
192-6605	71	Dwo	110	550	24	12	11		32.3x11.0x31.2
192-6606	71	W	110	550	24	12	11		32.3x11.0x31.2
192-6607	72	Dwo	110	660	24	12	13		31.2x12.9x31.2
192-6608	72	W	110	660	24	12	13		31.2x12.9x31.2
192-6609	73	Dwo	110	770	24	12	15		36.1x13.0x31.2
192-6610	73	W	110	770	24	12	15		36.1x13.0x31.2
192-6611	74	Dwo	110	880	24	12	17		40.6x13.0x31.2
192-6612	74	W	110	880	24	12	17		40.6x13.0x31.2
192-6613	75	Dwo	110	440	36	18	9		22.5x19.1x31.2
192-6614	75	W	110	440	36	18	9		22.5x19.1x31.2
192-7612	76	Dwo	110	550	36	18	11		30.1x18.0x31.2

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the current version before use.

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
192-7613	76	W	110	550	36	18	11		30.1x18.0x31.2
192-7614	77	Dwo	110	660	36	18	13		31.4x19.2x31.2
192-7615	77	W	110	660	36	18	13		31.4x19.2x31.2
192-7616	78	Dwo	110	770	36	18	15		31.9x24.2x31.2
192-7617	79	Dwo	110	880	36	18	17		38.5x20.6x31.2
192-7618	79	W	110	880	36	18	17		38.5x20.6x31.2
193-0523	34	W	85	595	12	6	15		36.1x7.8x23.5
193-0524	34	Dwo	85	595	12	6	15		36.1x7.8x23.5
193-0525	35	W	85	680	12	6	17		20.4x13.0x23.5
193-0526	35	Dwo	85	680	12	6	17		20.4x13.0x23.5
193-0527	36	W	85	765	12	6	19		38.4x7.8x23.5
193-0528	36	Dwo	85	765	12	6	19		38.4x7.8x23.5
193-0529	37	W	85	850	12	6	21		25.0x13.0x34.0
193-0530	37	Dwo	85	850	12	6	21		25.0x13.0x34.0
193-0531	38	W	85	935	12	6	23		38.4x9.2x23.5
193-0532	38	Dwo	85	935	12	6	23		38.4x9.2x23.5
193-0533	39	W	90	450	12	6	11		26.6x7.7x26.5
193-0534	39	Dwo	90	450	12	6	11		26.6x7.7x26.5
193-0535	47	Dwo	90	720	36	18	17		39.0x19.8x25.6
193-0536	48	W	90	810	36	18	19		38.4x22.3x25.6
193-0537	48	Dwo	90	810	36	18	19		38.4x22.3x25.6
193-0538	49	W	90	900	36	18	21		38.4x24.6x25.6
193-0539	50	Dwo	100	500	12	6	11		22.0x6.6x31.2
193-0540	51	W	100	600	12	6	13		31.2x7.7x23.5
193-0541	51	Dwo	100	600	12	6	13		31.2x7.7x23.5
193-0542	52	Dwo	100	700	12	6	15		36.1x7.7x31.2
193-0543	54	W	100	500	24	12	11		32.3x11.0x31.2
193-0544	54	Dwo	100	500	24	12	11		32.3x11.0x31.2
193-0545	40	W	90	540	12	6	13		31.2x7.7x26.5
193-0546	40	Dwo	90	540	12	6	13		31.2x7.7x26.5
193-0547	41	W	90	630	12	6	15		36.1x7.7x26.5
193-0548	41	Dwo	90	630	12	6	15		36.1x7.7x26.5
193-0549	42	W	90	720	12	6	17		20.4x13.0x26.5
193-0550	43	W	90	810	12	6	19		38.4x7.7x26.5
193-0551	43	Dwo	90	810	12	6	19		38.4x7.7x26.5
193-0552	44	W	90	450	36	18	11		26.1x19.3x25.6
193-0553	44	Dwo	90	450	36	18	11		26.1x19.3x25.6
193-0554	45	W	90	540	36	18	13		30.5x20.1x25.6
193-0555	45	Dwo	90	540	36	18	13		30.5x20.1x25.6
193-0556	46	W	90	630	36	18	15		35.3x19.4x25.6
193-0557	46	Dwo	90	630	36	18	15		35.3x19.4x25.6
193-0558	47	W	90	720	36	18	17		39.0x19.8x25.6
193-0559	55	W	100	600	24	12	13		31.2x12.9x31.2
193-0560	55	Dwo	100	600	24	12	13		31.2x12.9x31.2
193-0561	56	W	100	700	24	12	15		25.8x18.2x31.2
193-0562	56	Dwo	100	700	24	12	15		25.8x18.2x31.2
193-0563	57	W	100	800	24	12	17		27.3x19.3x31.2
193-0564	58	W	100	500	36	18	11		30.1x17.8x31.2

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L W H
193-0565	58	Dwo	100	500	36	18	11		30.1x19.3x31.2
193-0566	57	Dwo	100	800	24	12	17		27.3x19.3x31.2
193-2947	99	W	145	1160	36	18	17		41.5x27.0x31.2
193-2972	27	W	75	600	48	24	17		38.5x27.2x23.5
193-2973	95	W	145	580	36	18	9		26.6x22.7x31.2
193-2975	105	W	120	600	36	18	11	3x6	38.2x14.0x31.0
193-3704	7	W	75	600	12	6	17		20.5x13.0x23.5
193-3705	8	Dwo	75	675	12	6	19		22.7x13.0x23.5
193-3706	17	W	75	750	32	16	21		33.4x25.7x23.5
193-3707	18	W	75	825	32	16	23		36.4x25.7x23.5
193-3708	19	W	75	450	36	18	13		30.6x20.8x13.5
193-3709	19	Dwo	75	450	36	18	13		30.6x20.8x13.5
193-3710	20	W	75	525	36	18	15		31.9x24.2x23.5
193-3711	21	Dwo	75	600	36	18	17		33.8x25.7x23.5
193-3712	28	W	75	675	48	24	19		38.5x30.4x23.5
193-5070	80	Dwo	110	440	48	24	9		38.1x15.0x31.2
193-5071	80	W	110	440	48	24	9		38.1x15.0x31.2
193-5072	81	Dwo	110	550	48	24	11		38.1x18.0x31.2
193-5073	82	Dwo	110	660	48	24	13		38.1x21.0x31.2
193-5074	82	W	110	660	48	24	13		38.1x21.0x31.2
193-5075	83	Dwo	110	770	48	24	15		38.5x24.2x31.2
193-5076	83	W	110	770	48	24	15		38.5x24.2x31.2
193-5077	84	Dwo	110	880	48	24	17		40.7x25.7x31.2
193-6718	86	Dwo	145	725	12	6	11		27.0x9.1x31.2
193-6719	86	W	145	725	12	6	11		27.0x9.1x31.2
193-6720	87	Dwo	145	870	12	6	13		31.6x9.1x31.2
193-6721	88	Dwo	145	1015	12	6	15		36.1x9.1x31.2
193-6722	106	W	120	720	36	12	13	3x6	38.2x16.5x30.5
193-6723	49	Dwo	90	900	36	18	21		38.4x24.6x25.6
193-6724	53	W	100	800	12	6	17		38.4x7.7x31.2
193-6725	53	Dwo	100	800	12	6	17		38.4x7.7x31.2
193-6726	84	W	110	880	48	24	17		40.7x25.7x31.2
193-6727	85	Dwo	145	580	12	6	9		26.5x7.7x31.2
193-6728	85	W	145	580	12	6	9		26.5x7.7x31.2
193-7165	12	W	75	525	24	12	15		25.8x18.2x23.5
193-7166	14	Dwo	75	825	24	12	23		36.4x19.4x23.5
193-7167	124	W	85	850	48	24	21	4x6	38.7x33.5x22.6
193-7168	29	W	75	750	48	24	21		38.5x33.4x23.5
193-7169	113	W	85	680	36	18	17	3x6	34.2x26.5x23.6
193-7171	65	W	100	800	48	24	17		38.5x27.2x31.2
193-7190	59	W	100	600	36	18	13		30.6x20.8x31.2
193-8560	88	W	145	1015	12	6	15		36.1x9.1x31.2
193-8561	89	Dwo	145	1160	12	6	17		46.4x9.2x31.2
193-8562	89	W	145	1160	12	6	17		46.4x9.2x31.2
193-8563	90	Dwo	145	680	24	12	9		35.3x11.4x31.2
193-8564	90	W	145	680	24	12	9		35.3x11.4x31.2
193-8565	91	Dwo	145	725	24	12	11		35.3x13.6x31.2
193-8566	92	Dwo	145	870	24	12	13		35.5x16.1x31.2

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the current version before use.

NSN 6140-01-	Ind No.	Chg **	AH / + Pl	AH Cap 6 Hr R	V	No. of Cells	No. Plates	Cell * Layout	Dimensions L	W	H
193-8567	92	W	145	870	24	12	13		35.5	x16.1	x31.2
193-8568	93	Dwo	145	1015	24	12	15		36.2	x18.0	x31.2
193-8569	93	W	145	1015	24	12	15		36.2	x18.0	x31.2
193-8570	94	W	145	1160	24	12	17		36.2	x18.0	x31.2
194-0960	50	W	100	500	12	6	11		22.0	x6.6	x31.2
194-0961	52	W	100	700	12	6	15		36.1	x7.7	x31.2
194-0962	120	W	100	600	36	18	13	3x6	38.2	x16.5	x30.5
194-3207	81	W	100	500	48	24	11		38.1	x18.0	x26.5
194-3458	42	Dwo	90	720	12	6	17		20.4	x13.0	x26.5
194-5775	100	W	145	870	12	6	13	2x3	32.3	x9.2	x31.0
194-5776	101	W	85	595	12	6	15	2x3	18.7	x13.1	x23.2
195-0567	87	W	145	870	12	6	13		31.6	x9.1	x31.2
195-4575	23	Dwo	75	750	36	18	21		39.7	x25.7	x23.5
196-0087	91	W	145	725	24	12	11		35.3	x13.6	x31.2
196-0088	94	Dwo	145	1160	24	12	17		36.2	x18.0	x31.2
198-8355	78	W	110	770	36	18	15		31.9	x24.2	x31.2

*
cc - clipped corner
**
Dwe - Dry charged with electrolyte
Dwo - Dry charged without electrolyte
W - Wet charged

Source: https://assist.dla.mil -- Downloaded: 202610-26T11:25Z
Check the source to verify that this is the currentversion before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:25Z
Check the source to verify that this is the current version before use.